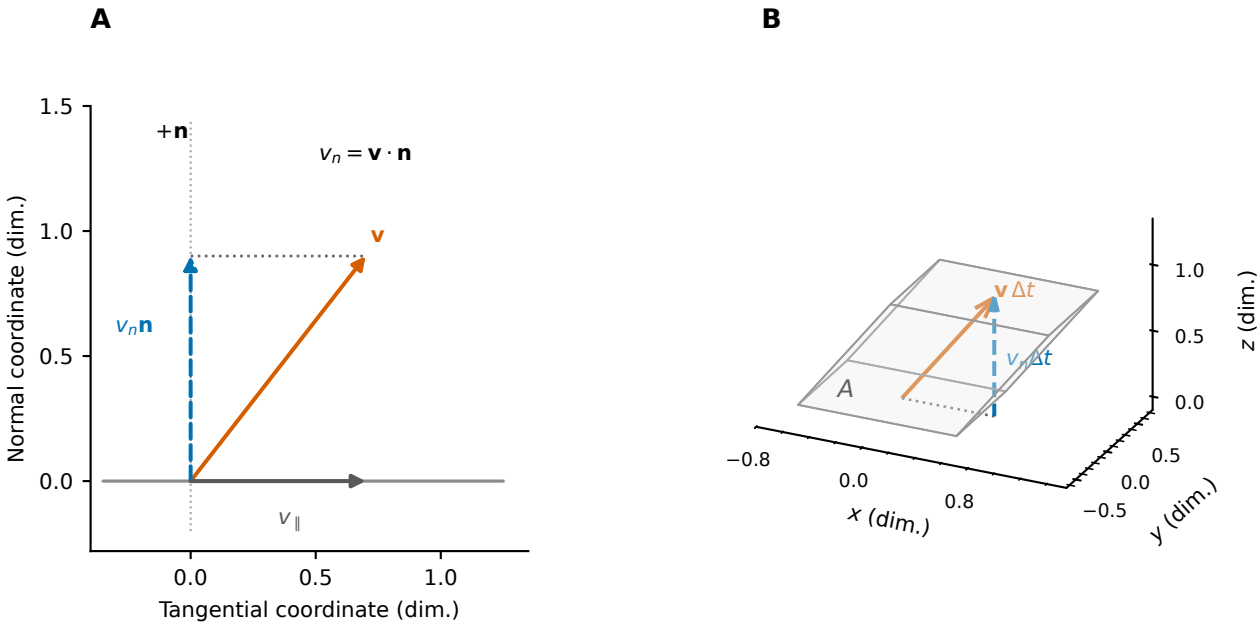

Only normal projection contributes: $\Phi = v_n A$

Swept volume: $\Delta V = A\, v_n \Delta t$